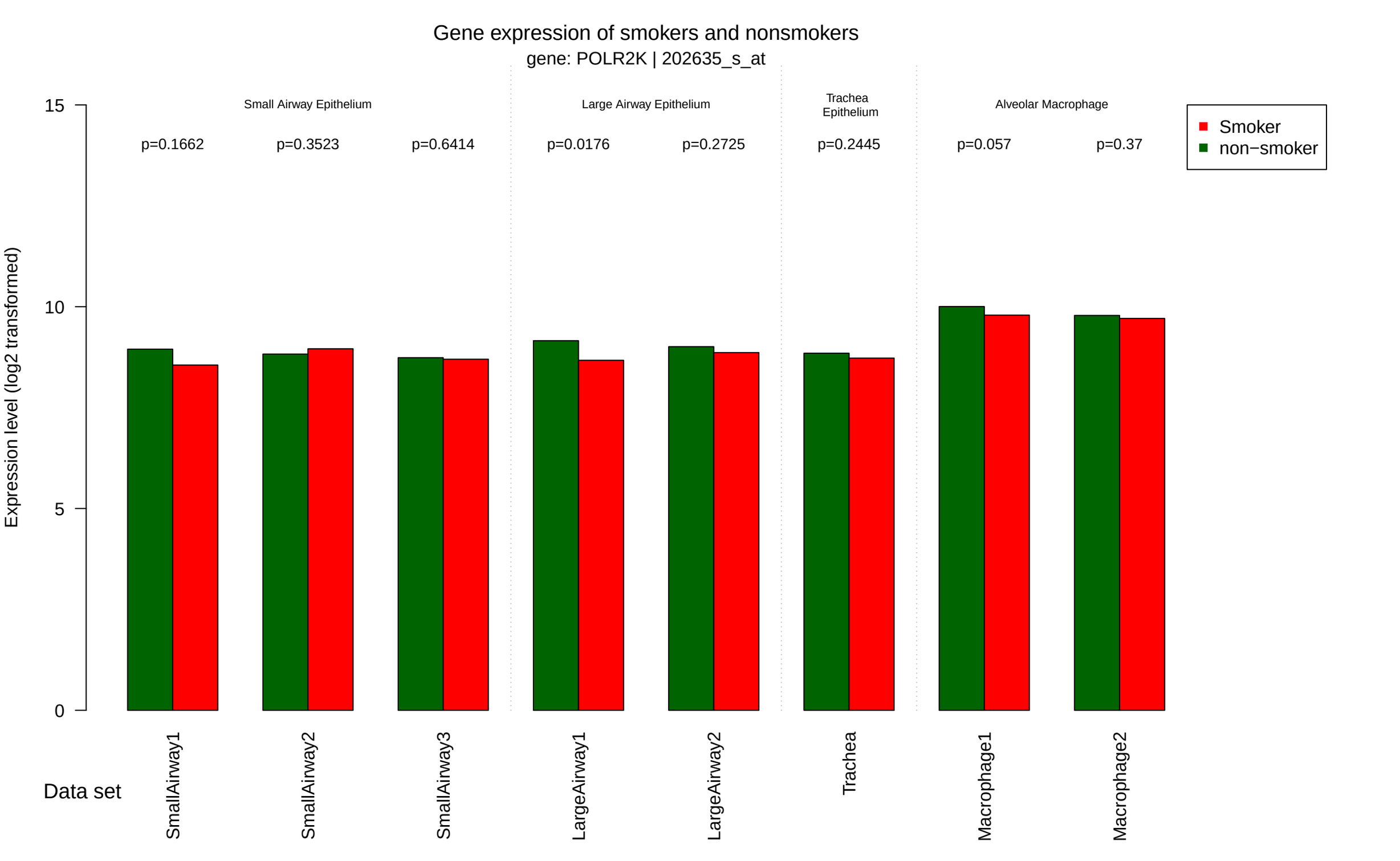
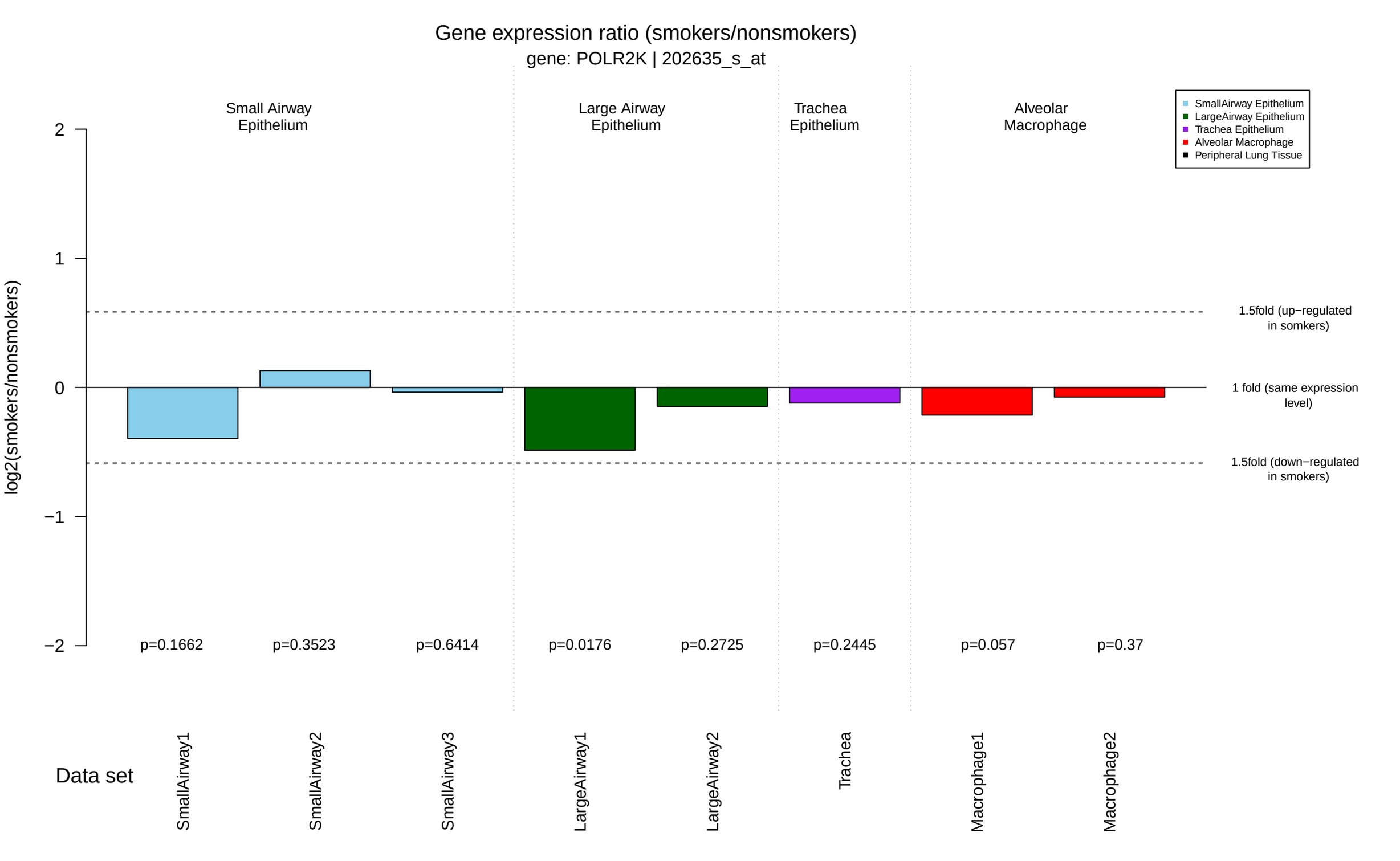
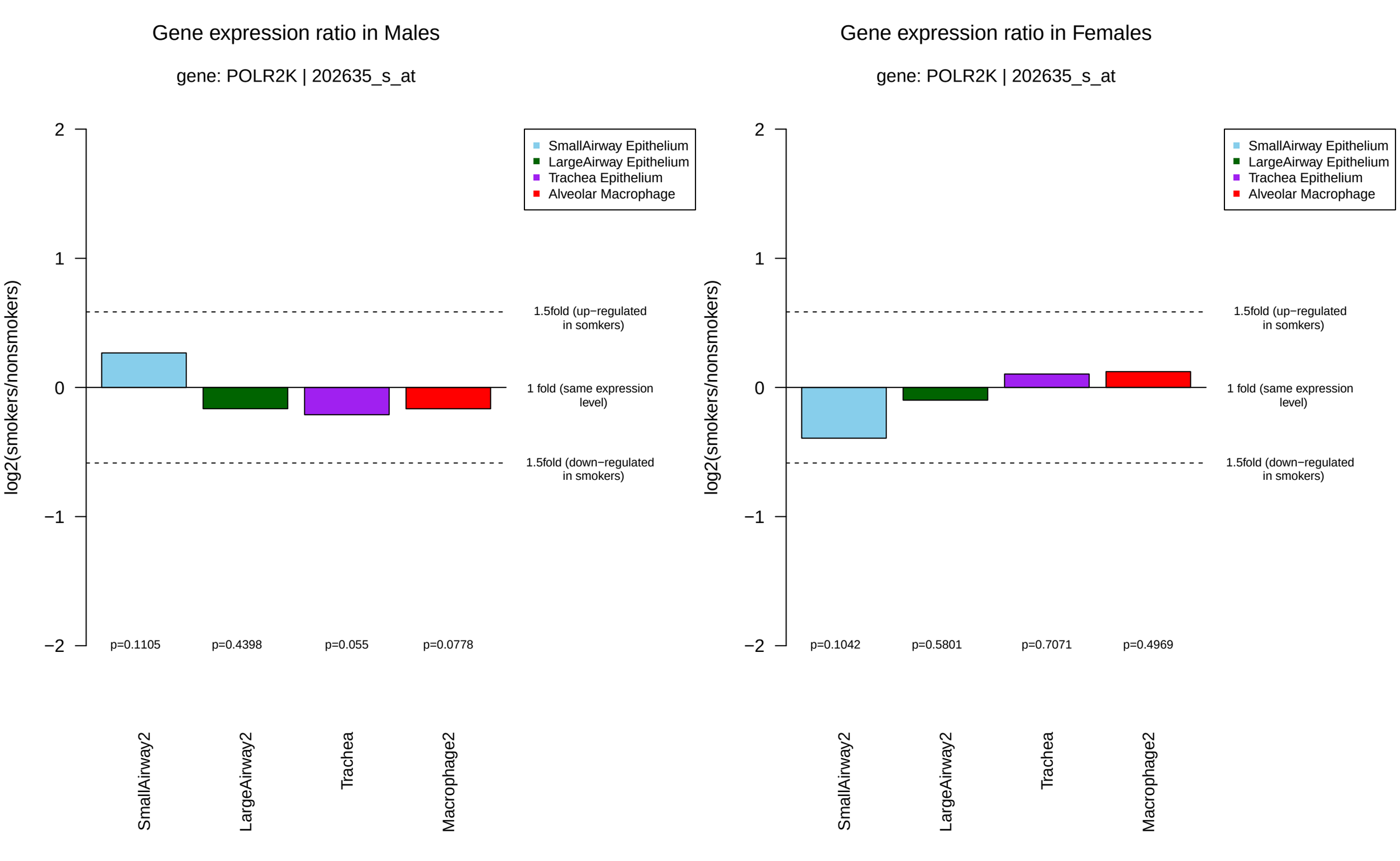
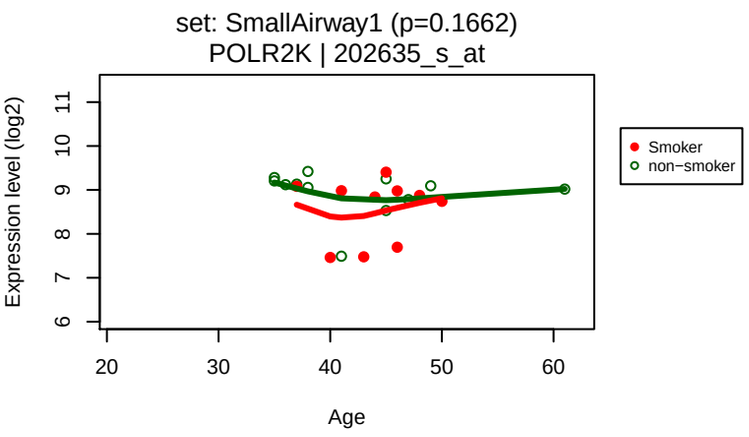
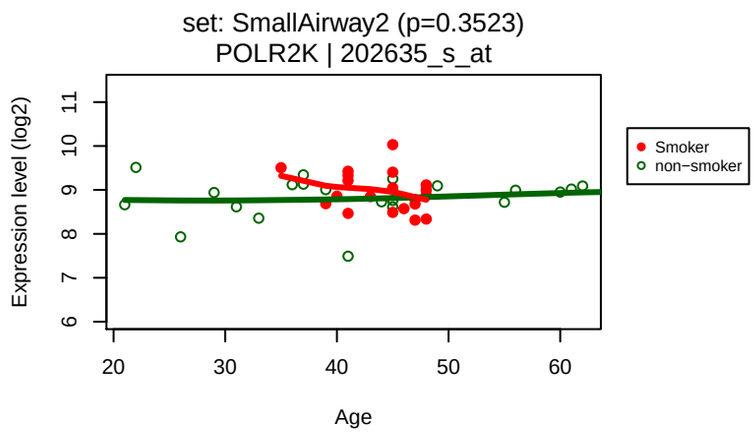
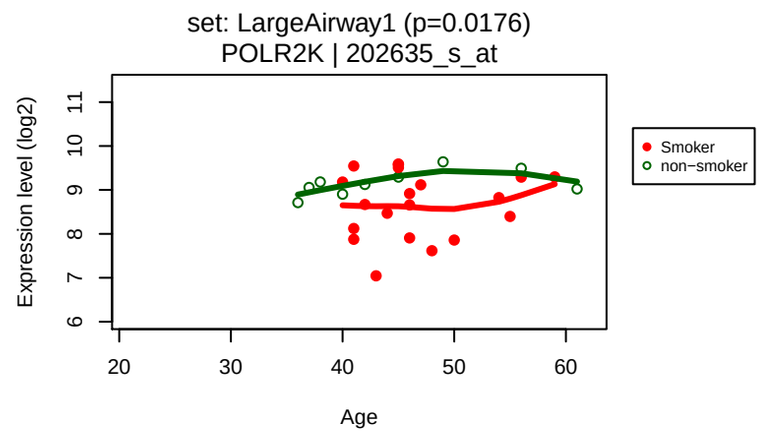

Gene expression of smokers and nonsmokers
gene: POLR2K | 202635_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1662
p=0.3523
p=0.6414
p=0.0176
p=0.2725
p=0.2445
p=0.057
p=0.37
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

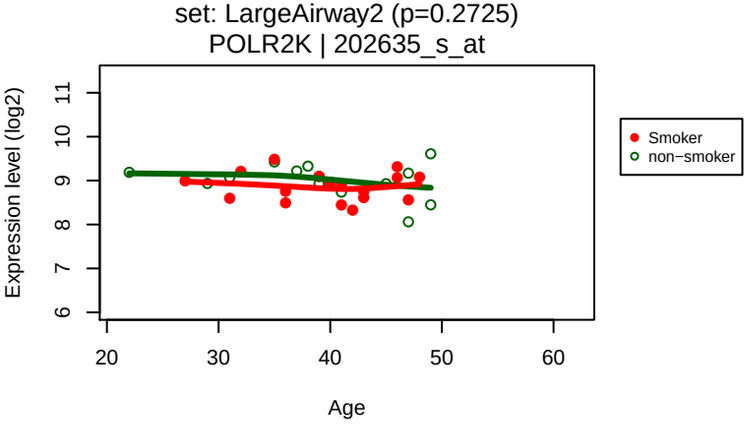

Gene expression ratio (smokers/nonsmokers)
gene: POLR2K | 202635_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1662
p=0.3523
p=0.6414
p=0.0176
p=0.2725
p=0.2445
p=0.057
p=0.37
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

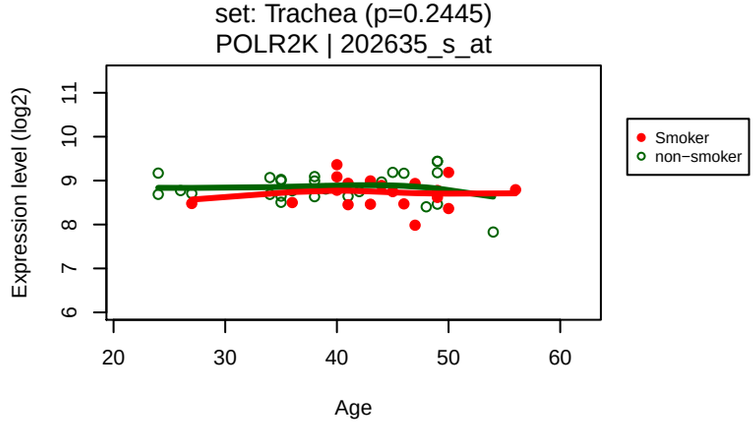

Gene expression ratio in Males
gene: POLR2K | 202635_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1105
p=0.4398
p=0.055
p=0.0778
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLR2K | 202635_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1042
p=0.5801
p=0.7071
p=0.4969
SmallAirway2
LargeAirway2
Trachea
Macrophage2

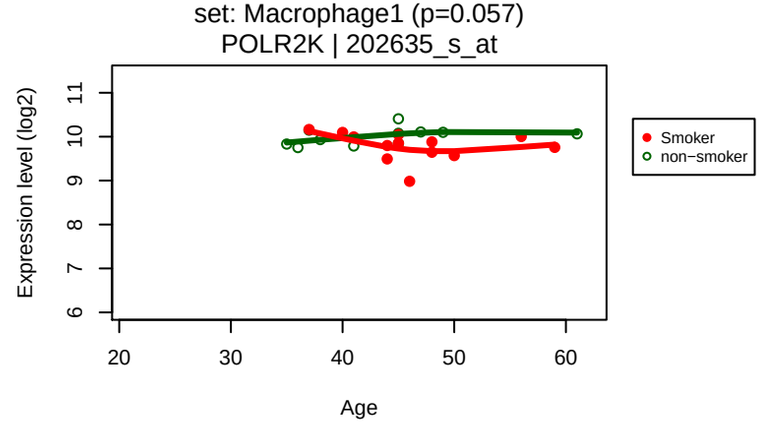

set: SmallAirway1 (p=0.1662)
POLR2K | 202635_s_at
Expression level (log2)
Age
Smoker
non−smoker

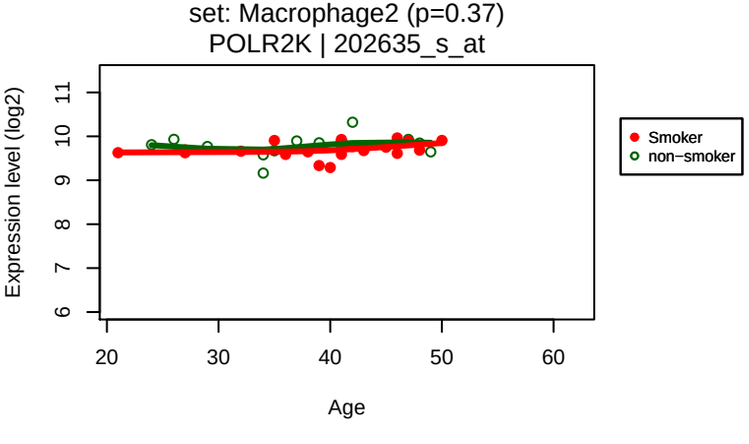

set: SmallAirway2 (p=0.3523)
POLR2K | 202635_s_at
Expression level (log2)
Age
Smoker
non−smoker

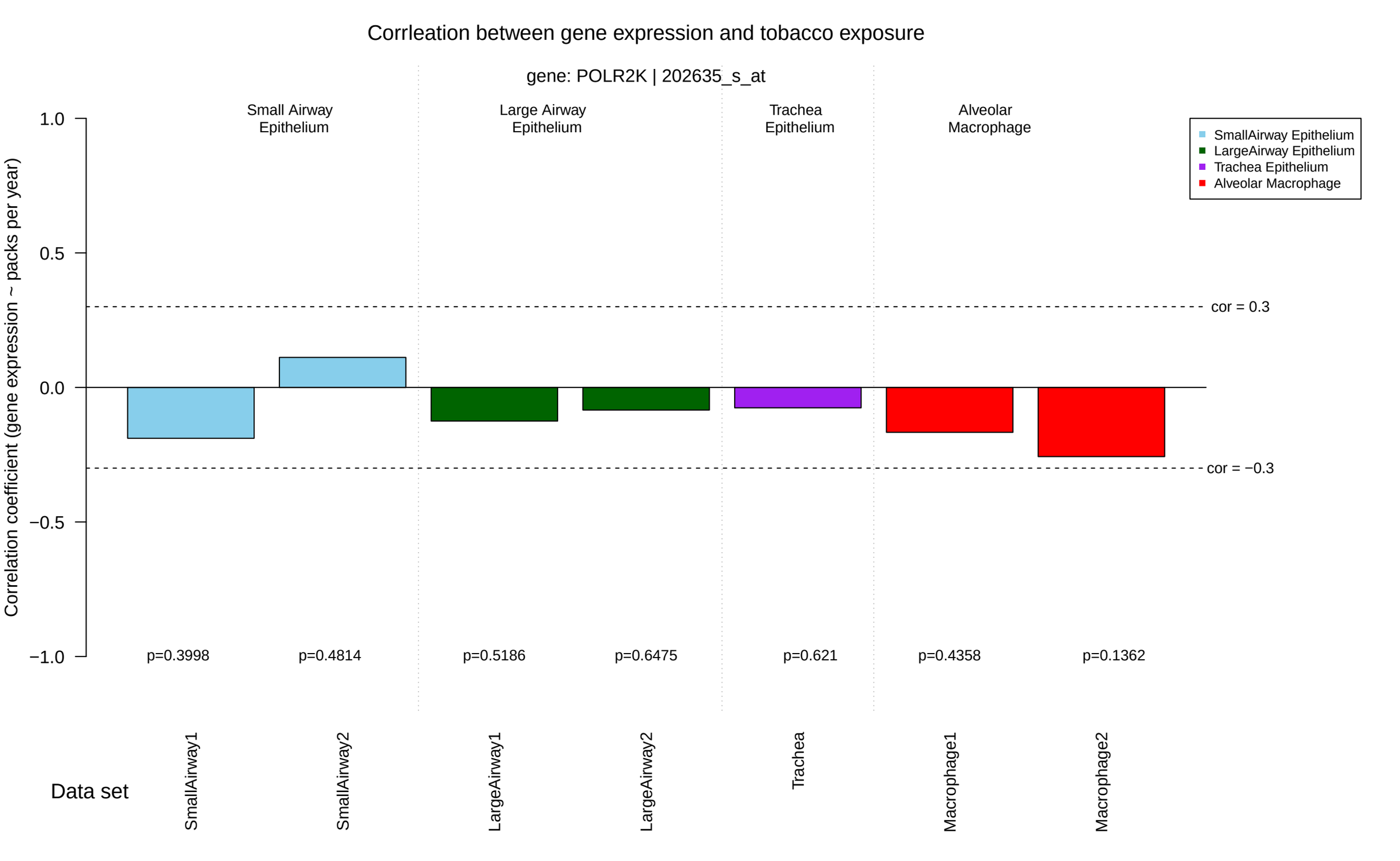

set: LargeAirway1 (p=0.0176)
POLR2K | 202635_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2725)
POLR2K | 202635_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2445)
POLR2K | 202635_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.057)
POLR2K | 202635_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.37)
POLR2K | 202635_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLR2K | 202635_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3998
p=0.4814
p=0.5186
p=0.6475
p=0.621
p=0.4358
p=0.1362
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2